

Primary Memory

Primary Memory

Where programs and data are stored

Can also be called *storage* but that term is usually reserved for **disk storage**

A non-negotiable part of computer operation

Bits

Bits

The basic unit of memory for all computers is the **bit**

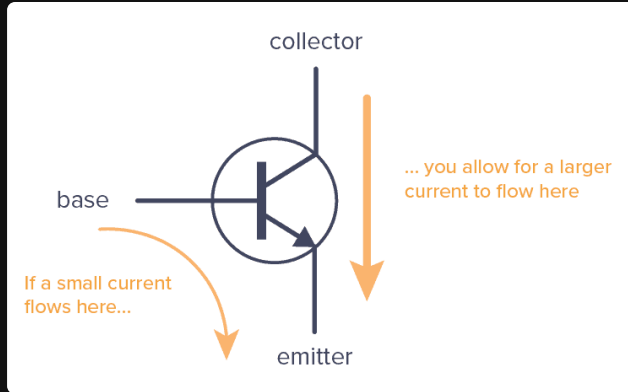
A bit contains either a **1** or a **0**

And computers use bits because they are more "*efficient*"

Specifically, bits offer two main benefits

- reliability
- information density

Reliability



A transistor works by being an *on* and *off* switch

There are 3 pins

- *control*,
- *input*, and
- *output*

By applying a *small voltage* to the control pin, it will **open or close** the circuit,

This allows the input to go to the output

And if the output is over a certain *threshold*

it's considered a 1

If we made the threshold smaller, we could get more values `0 1 2 3 4`

However, the likelihood of *noise* affecting the reading increases

Information Density

Some older large computers are advertised as having *decimal arithmetic*, they do this using **4 bits** to store a single decimal digit called a **BCD** (Binary Coded Decimal)

Let's assume **16** bits

Both of the examples below show *1944*, but they have different **maximums**

BCD

```
1  0001 1001 0100 0100
```

Showcases 4 digits that can start at **0** and end at **9,999**

```
1  1001 1001 1001 1001
```

For a total of **10,000** unique combinations

Binary

```
1  0000 0111 1001 1000
```

Showcases 16 digits of binary that can have a maximum of **65,535**

And can store a total of **65,536** unique combinations

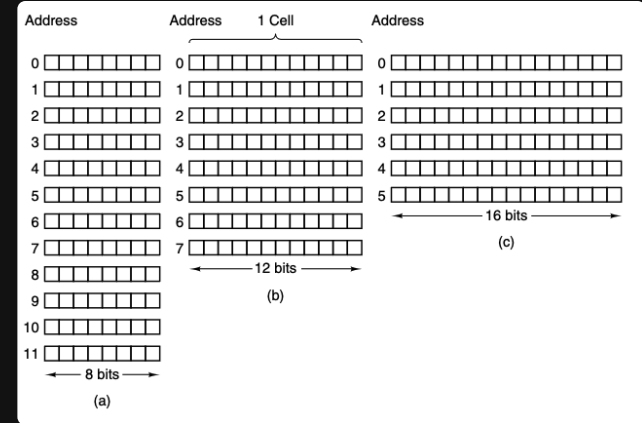
Memory Addresses

Memory Addresses

Memories consist of a number of **cells**

- each of which can store a *piece of information*,
- each has a number, called its **address** which programs can refer to,
- if a memory has n cells, they will have *addresses* 0 to $n - 1$, and
- all cells in a memory contain the *same number of bits*
- if a cell consists of k bits, then it can hold 2^k *combinations*

Below is an example of different combinations of a **96-bit** memory



Memory Addresses (cont)

To **address** (meaning to find), a computer needs to have a certain amount of bits *reserved*

If an address has m bits, the maximum number of cells *addressable* is 2^m

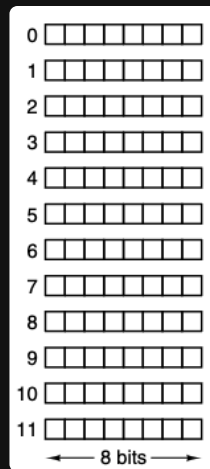
For example

To be able to address *all* the memory locations on the example on the right

We need a **minimum length of 4 bits**

Because an address length of 3 bits can only address up to $2^3 = 8$ bits

Which cell does `0100` address?



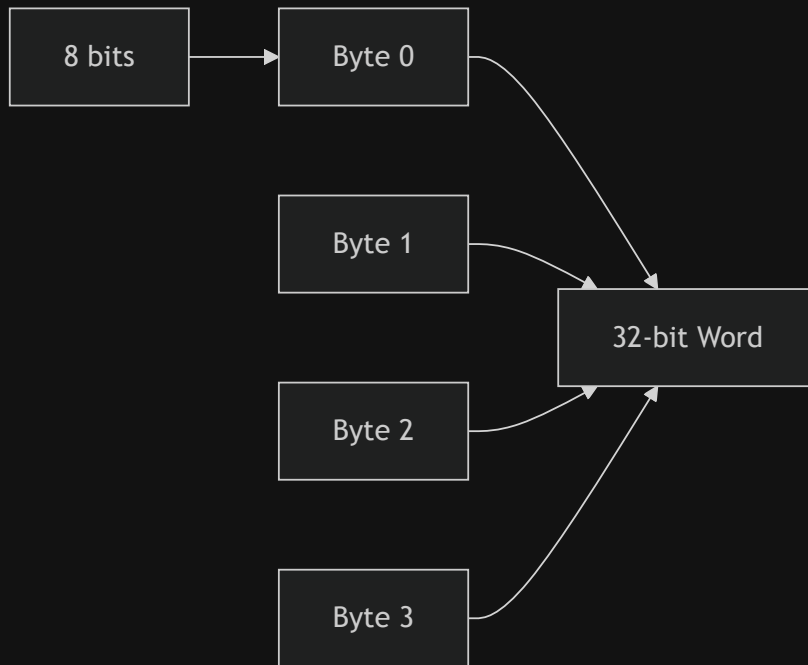
Cell

A **Cell** is the smallest *addressable unit* (row)

In recent years, the majority of computer manufacturers have standardized on an *8-bit cell*, which is called a **byte** (sometimes *octet*)

bytes are then grouped into **words**

- A 32-bit computer has 4 bytes/word
- A 64-bit computer has 8 bytes/word



Bits vs Bytes

Storage and networking are sold in **bits** or **bytes**

Scenario	Advertised as	What you get
Internet speed	500 Mbps (megabits/sec)	62.5 MB/s (megabytes/sec) - 8x slower
SSD read speed	3500 MB/s	same - storage uses <i>bytes</i>
RAM bandwidth	25.6 GB/s	same - also bytes
Network interface	10 Gbps	1.25 GB/s real transfer

A 500 Mbps plan does **not** download a 5 GB file in 80 seconds.

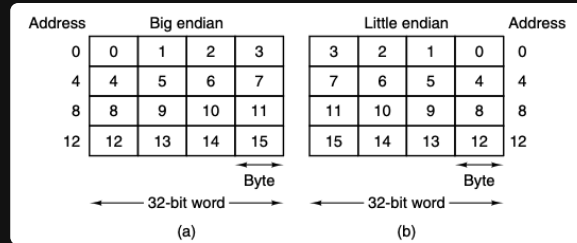
It downloads 5 Gb (gigabits) in 80 seconds – which is only 0.625 GB .

Byte Ordering

Byte Ordering

Bytes in a word can be numbered *left to right* or *right to left*

This is *relevant*



- *a* represents a *left to right* ordering of a 32-bit computer
- *b* represents a *right to left* ordering

a would be considered a **big endian** because the numbers start at the *big end (higher order)* and vice versa for **little endian**

Byte Ordering (cont)

For example,

Given a 32-bit integer of value 6 is represented by the bits 110 in the rightmost (*low-order*) 3 bits of a word

```
1  00000000 00000000 00000000 00000110
2  0          1          2          3
```

In a *big endian scheme*, the 110 bits would be in byte 3 (or 7, or 11, etc)

In a *little endian scheme*, the 110 bits would be in byte 0

In **both** cases, the *word* that contains this integer has the address of 0

Byte Ordering (cont)

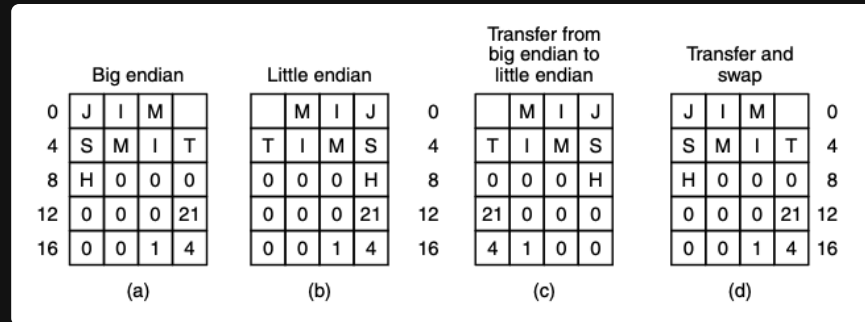
If computers *only stored integers*, there would be no problem

But computers use a *mixture* of integers, characters, and other data types

For example,

A simple personnel record (name `string`, age `int`, dept_id `int`)

And the string is terminated by `1` or more `0` bytes to fill out a word



Where *a* is the *big endian* representation, and *b* is the *little endian* representation

Byte Ordering (cont)

Big endian				Little endian				Transfer from big endian to little endian				Transfer and swap			
0	J	I	M			M	I	J	0			M	I	J	0
4	S	M	I	T		T	I	M	S	4		T	I	M	S
8	H	0	0	0		0	0	0	H	8		0	0	0	H
12	0	0	0	21		0	0	0	21	12		21	0	0	0
16	0	0	1	4		0	0	1	4	16		4	1	0	0
(a)				(b)				(c)				(d)			

A problem arises when one wants to **send the record** to another machine

If we assume that the *big endian* sends a record to the *little endian* **one byte at a time**, starting at byte 0 and ending at byte 19

assuming that the bits of the byte aren't reversed

So the big endians byte 0 goes into the little endians byte 0

Printing the name would be fine, but the age comes out as $21 \cdot 2^{24}$ and the department is garbled

Byte Ordering (cont)

In the modern day, this is *largely* fixed thanks to the separation of formatting of data and cpu reading

1. Text-based Serialization

We normally send data using text like `JSON` which has a more standard way of decoding (UTF-8) which is byte-order independent

2. Binary Serialization Frameworks

Things like Protobufs which both the sender and the receiver *agree upon* beforehand and so will correctly flip bytes as required

3. Network Byte Order

`TCP/IP` has largely agreed that everyone should be using *big-endian* (at least for low-level networking)

4. Little Endian Monoculture

Little-Endian has largely won the *consumer hardware war*.

Error Correcting Codes

Error Correcting Codes

Electricity and voltage are not reliably *stable*, voltage spikes, cosmic rays, another appliance being plugged in

Super mario speedrun helped by a space ray

All of these can cause memory to **error**, which is why *error correcting codes* exist

Similar to how a QR has *more data than required*, extra bits are added to each memory word in a special way to

1. **check** if an error has occurred
2. **fix** that error if possible

What is an error

Suppose you have a memory word consisting of m data bits

We will add r redundant, or *check*, bits

So the total length is $n = m + r$

An n -bit unit containing m data and r check bits is often referred to as a

Codeword

What is an error (cont)

Given any two codewords

```
1  1000 1001 and
2  1011 0001
```

It is possible to determine how many corresponding bits **differ** using *exclusive or*

```
1  1000 1001 xor
2  1011 0001
3  _____
4  0011 1000
```

And simply counting how many 1 bits are in the result

This is called **Hamming distance**

Hamming distance

if we assume that two codewords are d hamming distance apart, it will *require* d single bit errors to convert one into another

for example

```
1  1111 0001
2  0011 0000
3  _____
4  1100 0001
```

3 hamming distance apart, meaning it will take **3 errors** to convert *one into another*

Error detection

if we assume

- 3 m bits (data)
- 2 r bits (check)
- $n = 5$ bits (total)

all 2^m bit patterns are legal (2^3)

1	000
2	001
3	010
4	011
5	100
6	101
7	110
8	111

but with check bits, because of how they are *computed*, only 2^m of the 2^n bits are **valid**

	data	check	valid
1			
2	000	00	00000
3	001	01	00101

But

1	data	check	invalid
2	001	00	00100

So if a memory read turns out an *invalid codeword*, the computer **knows** that a memory error has occurred

Error detection

Given an algorithm for *computing check* bits, it is possible to construct a *complete list of the legal codewords*, and from this list

we can

1. find the two code words who have the smallest hamming distance

And this distance is the **hamming distance of the complete code**

The ability for a code to *detect* and *correct* errors **depend** on this hamming distance

Error correction

To *detect* d single bit errors, you need a *minimum distance* of $d + 1$ to another valid codeword

Because if you have a code like that,

- the *only way* a legal codeword can change *into another* legal codeword
- is with $> d$ errors

And if you want to correct a code,

You'd need $2d + 1$ distance, so that all legal codewords are so far apart that even d changes

The original codeword is still *closer* than any other codeword

Parity bit

A simple example of an error detecting code is a **parity bit**

The parity bit is chosen so that the *number of 1 bits* in the codeword is *even* (or odd)

This code has a **distance** of **2**

Because it takes *two single bit errors* to go from a valid codeword to another valid codeword

So if it detects a word containing the wrong parity when pulling from memory, **an error happens**

The program **can't** continue but no incorrect results are computed

Error Correction

Consider a code with only four valid codewords

```
1  00000 00000
2  00000 11111
3  11111 00000
4  11111 11111
```

This code has a distance of 5

The minimum amount of errors it would take to turn one codeword into another is 5

Which means it can **correct** double errors ($(2 * 2) + 1 = 5$)

So if `00000 00111` arrives, the receiver knows that the original **must be** `00000 11111` since that's the *closest*

But if we had a **3 bit error** so `00000 00000` turns into `00000 00111` we can't correct it

Scenario

Say we want to *design a code* with m data bits and r check bits that will allow **all single bit errors** to be corrected

So each of the 2^m legal memory words has n *illegal codewords* at a **distance 1** from it

Thus each of the 2^m legal memory words require $n + 1$ bit patterns dedicated to it (1 correct, and 1 for every bit being flipped)

```
1  Given 000
2
3  001
4  010
5  100
```

Scenario

We know that there are 2^m **legal data words**

If every one of those 2^m words requires its *own unique neighbor* of $n + 1$ patterns

We need

$(n + 1)2^m$ bits

Scenario

Then, since a memory slot of n bits can only hold a *maximum* of 2^n **unique combinations**

Our total required "*reserved*" patterns cannot exceed the total number of physically available patterns

$$(n + 1)2^m \leq 2^n$$

Since we know that the total codeword length is $n = m + r$

we can substitute

$$\begin{aligned}(m + r + 1)2^m &\leq 2^{m+r} \\ (m + r + 1)2^m &\leq 2^m \cdot 2^r \\ m + r + 1 &\leq 2^r\end{aligned}$$

This is saying that

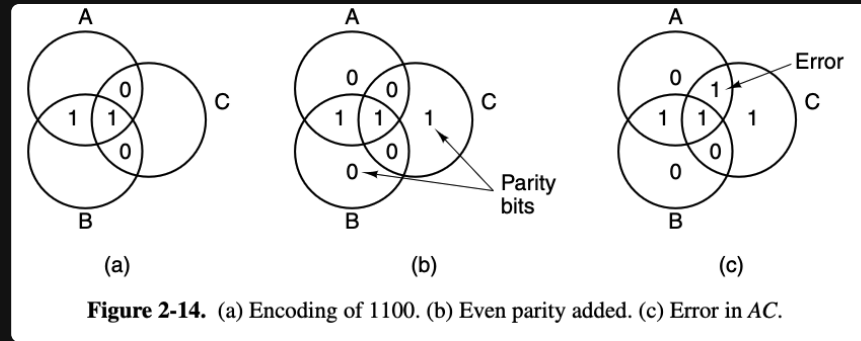
If we choose a **data word** size m , we can find the absolute **minimum number of check bits** (r) required to achieve single error correction

Scenario

Word size	Check bits	Total size	Percent overhead
8	4	12	50%
16	5	21	31%
32	6	38	19%
64	7	71	11%
128	8	136	6%
256	9	265	4%
512	10	522	2%

Visual

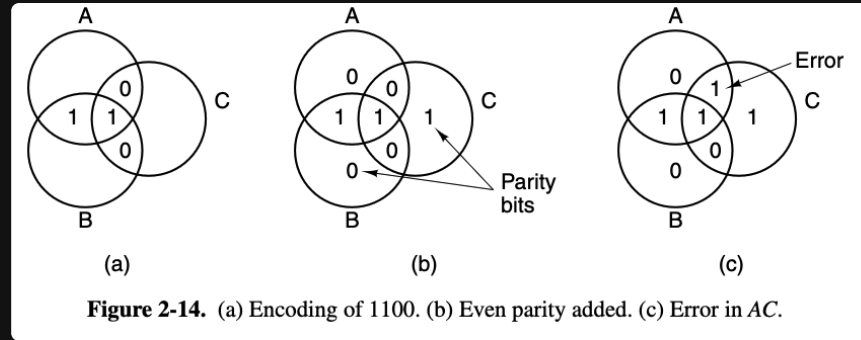
Let's look at the error correcting code for 4 bit words



If we encode **1100** in the regions AB , ABC , AC and BC , with 1 bit per region (alphabetical order)

We get figure *a*

Visual

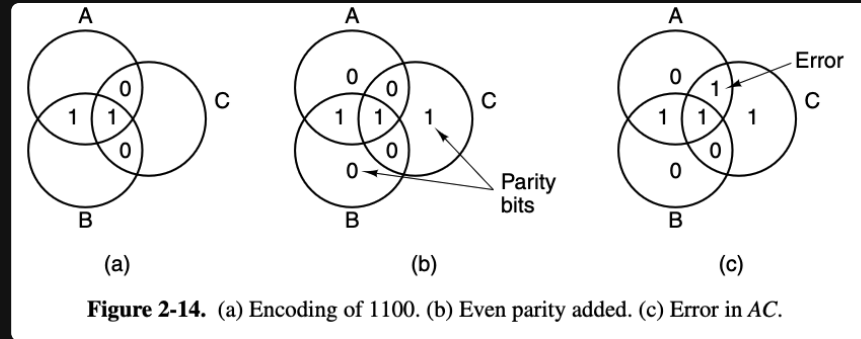


Then we add a **parity** bit to each of the three empty regions to produce **even parity** (b)

So, the sum of the bits in each of the three circles *A*, *B*, and *C* is now an *even number*

This figure corresponds to a codeword with 4 *data bits* and 3 *parity bits*

Visual



If the region in AC goes bad (0 to 1), the computer can now see that circles A and C have the *wrong parity*

The **only** single bit change that corrects them is to restore AC back to 0 thus **correcting the error**

So single bit errors **repair automatically**

Hamming's algorithm

We can use Hamming's algorithm to construct *error correcting codes* for any size memory word

In a Hamming code r **parity bits** are added to an m bit word forming $m + r$ bits

The bits are numbered starting at 1 with bit 1 being the *left most*

```
1  0 0 0 0 0 0 0 0 0
2  ^ parity bit 1
```

All bits whose number is a *power of 2* are parity bits, the rest are data

```
1  0 0 0 0 0 0 0 0 0
2  ^ ^ ^      ^ parity bit 1, 2, 4, and 8
```

For example, in a 16-bit word, **5 parity bits** would be added

1 2 4 8 and 16 will become *parity* bits, the rest are **data**

So the memory word has **21**-bits (16 data 5 parity)

In an even parity example

Each parity bit **checks specific bit positions**, the parity bit is set so that the total number of 1s in the checked positions is even

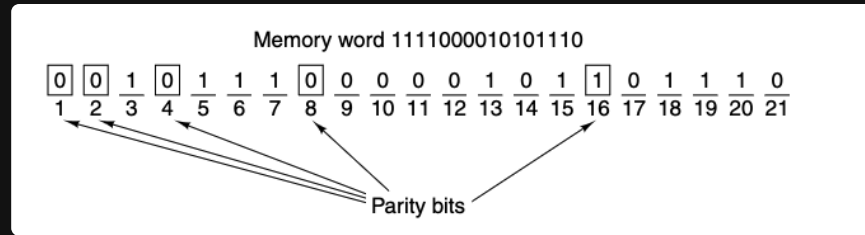
- 1 bit 1 checks bits 1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21
- 2 bit 2 checks bits 2, 3, 6, 7, 10, 11, 14, 15, 18, 19
- 3 bit 4 checks bits 4, 5, 6, 7, 12, 13, 14, 15, 20, 21
- 4 bit 8 checks bits 8, 9, 10, 11, 12, 13, 14, 15
- 5 bit 16 checks bits 16, 17, 18, 19, 20, 21

In general, *each bit* b is checked by those bits $b_1, b_2, \dots, b_j$ such that $b_1 + b_2 + \dots + b_j = b$

So because 11 is $11 = 8 + 2 + 1$, it will be **checked** by parity bit 8, 2, and 1

In an even parity example

Given a Hamming code for 16 bit memory with a 21 bit codeword



Consider if the 5th bit was inverted

- 0010 1110 0000 1011 0111 0 original
- 0010 0110 0000 1011 0111 0 error

In an even parity example

```
1  0010 1110 0000 1011 0111 0 original
2  0010 0110 0000 1011 0111 0 error
```

```
1  Parity bit 1   incorrect (1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21 contain five 1s).
2  Parity bit 2   correct   (2, 3, 6, 7, 10, 11, 14, 15, 18, 19 contain six 1s).
3  Parity bit 4   incorrect (4, 5, 6, 7, 12, 13, 14, 15, 20, 21 contain five 1s).
4  Parity bit 8   correct   (8, 9, 10, 11, 12, 13, 14, 15 contain two 1s).
5  Parity bit 16  correct   (16, 17, 18, 19, 20, 21 contain four 1s).
```

To **find** the incorrect bits,

- first **compute all the parity bits**
- then **add up** all incorrect parity bits, starting at 1

$$1 + 4 = 5$$

In an even parity example

```
1  0010 1110 0000 1011 0111 0 original
2  0010 1110 0100 1011 0111 0 error
```

```
1  Parity bit 1  _____ (1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21 contain _ 1s )
2  Parity bit 2  _____ (2, 3, 6, 7, 10, 11, 14, 15, 18, 19 contain _ 1s )
3  Parity bit 4  _____ (4, 5, 6, 7, 12, 13, 14, 15, 20, 21 contain _ 1s )
4  Parity bit 8  _____ (8, 9, 10, 11, 12, 13, 14, 15 contain _ 1s )
5  Parity bit 16 _____ (16, 17, 18, 19, 20, 21 contain _ 1s )
```